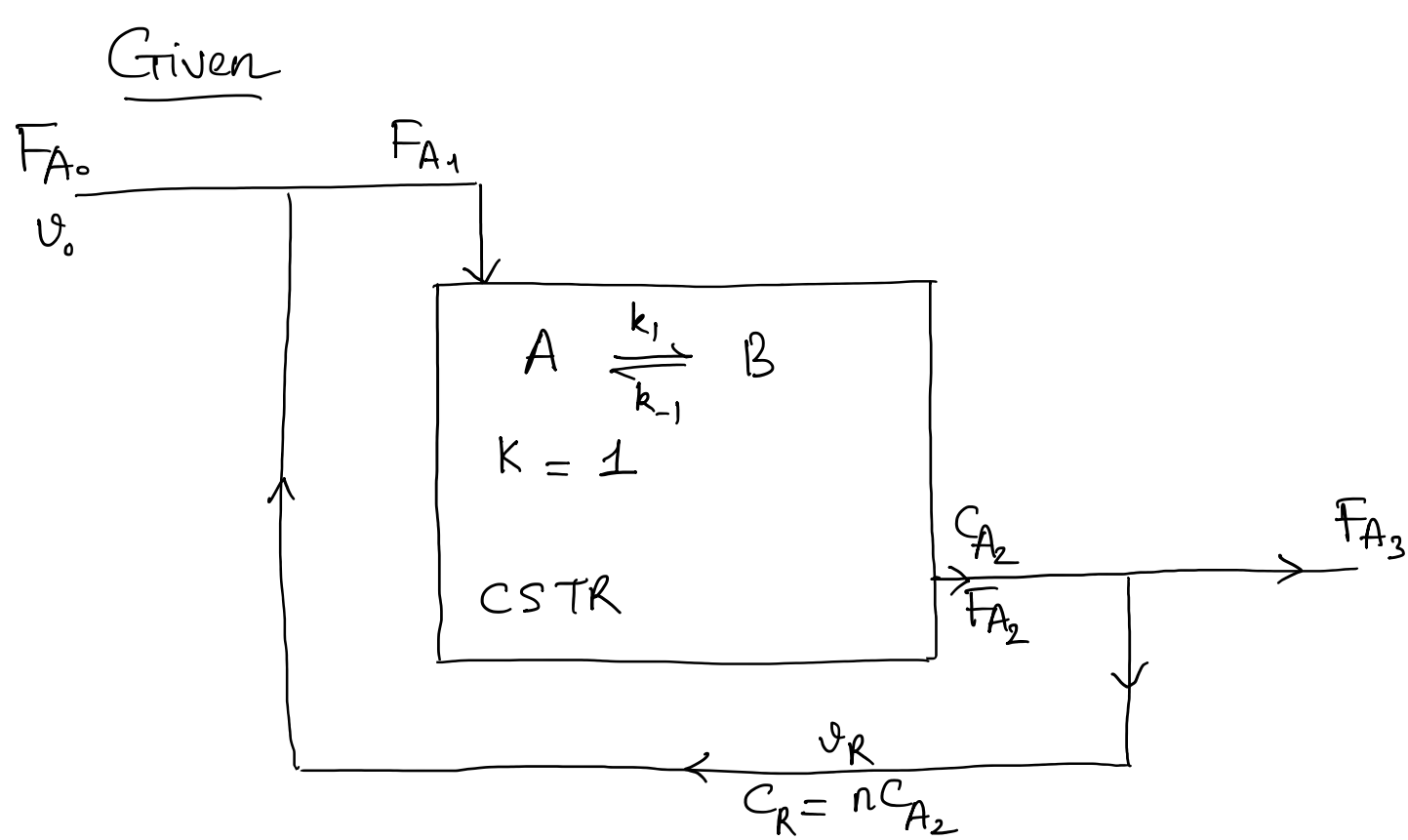




$$\bullet \text{ Total Conversion } X_T = \frac{F_{A_0} - F_{A_3}}{F_{A_0}} = 1 - \frac{F_{A_3}}{F_{A_0}}$$

$$\bullet \alpha \equiv \frac{\text{molar flow rate recycled}}{\text{total molar flow rate coming out from CSTR}}$$

$$\alpha = \frac{F_R}{F_{A_2}} \quad (\text{by definition})$$

$$\alpha = \frac{F_R}{F_{A_2}} = \frac{C_R v_R}{C_{A_2} v_2} = \frac{n C_{A_2} v_R}{C_{A_2} v_2} = \frac{n v_R}{v_2} = \frac{n v_R}{v_2}$$

$$(1) \bullet F_{A_1} = F_{A_0} + F_R = F_{A_0} + \alpha F_{A_2}$$

$$(2) \bullet F_{A_2} = F_{A_1} (1 - X) \quad ; \quad F_{A_1} - F_{A_2} = F_{A_1} X \quad \langle X \neq X_T \rangle$$

$$\Rightarrow F_{A_1} = F_{A_0} + \alpha F_{A_1} (1 - X)$$

$$\Rightarrow F_{A_0} = F_{A_1} - \alpha F_{A_1} (1 - X)$$

$$= F_{A_1} [1 - \alpha (1 - X)]$$

$$\Rightarrow F_{A_1} = \frac{F_{A_0}}{1 - \alpha (1 - X)}$$

$$\omega \Rightarrow F_{A_1} - F_{A_2} = F_{A_1} X = \frac{F_{A_0} X}{1 - \alpha (1 - X)}$$

⊛ CSTR balance

$$\bullet F_{A_1} - F_{A_2} + R_A V = 0$$

$$\Rightarrow \frac{F_{A_0} X}{1 - \alpha (1 - X)} - (k_1 C_{A_2} - k_{-1} C_{B_2}) V = 0$$

$$\text{We know } C_{A_2} = C_{A_1} (1 - X) = \frac{F_{A_1}}{v_1} (1 - X)$$

$$C_{B_2} = C_{A_1} X$$

$$\Rightarrow \frac{F_{A_0} X}{1 - \alpha (1 - X)} - \left[\frac{k_1 F_{A_1} (1 - X)}{v_1} - \frac{k_{-1} F_{A_1} X}{v_1} \right] V = 0$$

$$\text{Because } K = 1 \Rightarrow k_1 = k_{-1}$$

$$\Rightarrow \frac{F_{A_0} X}{1 - \alpha (1 - X)} - \frac{V}{v_1} k_1 F_{A_1} (1 - 2X) = 0$$

$$\Rightarrow \frac{F_{A_0} X}{1 - \alpha (1 - X)} - \frac{V k_1}{v_1} \frac{F_{A_0}}{1 - \alpha (1 - X)} (1 - 2X) = 0$$

$$\Rightarrow \frac{F_{A_0}}{1 - \alpha (1 - X)} \left[X - \frac{V k_1}{v_1} (1 - 2X) \right] = 0$$

$$\Rightarrow X - \frac{V k_1}{v_1} (1 - 2X) = 0$$

$$\Rightarrow X - \frac{V k_1}{v_1} + \frac{2X V k_1}{v_1} = 0$$

$$\Rightarrow X \left(1 + \frac{2V k_1}{v_1} \right) = \frac{V k_1}{v_1}$$

$$\Rightarrow X \left(\frac{v_1 + 2V k_1}{v_1} \right) = \frac{V k_1}{v_1}$$

$$\Rightarrow X = \frac{V k_1}{v_1 + 2V k_1} = \frac{V k_1}{v_0 + v_R + 2V k_1} \quad +50$$

$$\bullet F_{A_2} = F_{A_3} + F_R = F_{A_3} + \alpha F_{A_2}$$

$$\Rightarrow F_{A_3} = F_{A_2} (1 - \alpha) \quad \left\{ \Rightarrow F_{A_3} = F_{A_1} (1 - X) (1 - \alpha) \right.$$

$$\bullet F_{A_2} = F_{A_1} (1 - X)$$

$$\bullet F_{A_0} = F_{A_1} - F_R = F_{A_1} - \alpha F_{A_2}$$

$$= F_{A_1} - \alpha F_{A_1} (1 - X)$$

$$= F_{A_1} [1 - \alpha (1 - X)]$$

$$= \frac{F_{A_3}}{(1 - X)(1 - \alpha)} [1 - \alpha (1 - X)]$$

$$\Rightarrow \frac{F_{A_3}}{F_{A_0}} = \frac{(1 - X)(1 - \alpha)}{1 - \alpha (1 - X)}$$

$$\Rightarrow -X_T = \frac{F_{A_3}}{F_{A_0}} - 1$$

$$= \frac{(1 - X)(1 - \alpha)}{1 - \alpha (1 - X)} - \frac{1 - \alpha (1 - X)}{1 - \alpha (1 - X)}$$

$$= \frac{\cancel{1} - X - \cancel{\alpha} + X\cancel{\alpha} - \cancel{1} + \cancel{\alpha} - X\cancel{\alpha}}{1 - \alpha (1 - X)}$$

$$= \frac{-X}{1 - \alpha (1 - X)}$$

$$\Rightarrow X_T = \frac{X}{1 - \alpha (1 - X)}$$

$$X = \frac{V k}{v_0 + v_R + 2V k} \quad ; \quad 1 - X = \frac{v_0 + v_R + 2V k - V k}{v_0 + v_R + 2V k} = \frac{v_0 + v_R + V k}{v_0 + v_R + 2V k}$$

$$= \frac{\frac{V k}{v_0 + v_R + 2V k}}{1 - \frac{n v_R}{v_0 + v_R} \cdot \frac{v_0 + v_R + V k}{v_0 + v_R + 2V k}}$$

$$\alpha = \frac{n v_R}{v_0 + v_R}$$

$$\frac{V k}{v_0 + v_R + 2V k} \cdot \left[\frac{(v_0 + v_R)(v_0 + v_R + 2V k)}{(v_0 + v_R)(v_0 + v_R + 2V k)} - \frac{n v_R (v_0 + v_R + V k)}{(v_0 + v_R)(v_0 + v_R + 2V k)} \right] =$$

$$\frac{V k}{v_0 + v_R + 2V k} \times \frac{(v_0 + v_R)(v_0 + v_R + 2V k)}{(v_0 + v_R)(v_0 + v_R + 2V k) - n v_R (v_0 + v_R + V k)} =$$

$$\frac{V k (v_0 + v_R)}{(v_0 + v_R)(v_0 + v_R + 2V k) - n v_R (v_0 + v_R + V k)} = X_T$$

$$b) \quad v_R = 0 \Rightarrow X_T = \frac{V k v_0}{v_0 (v_0 + 2V k)}$$

$$= \frac{V k}{v_0 + 2V k} \quad +20$$